

The medicine is already there. Nobody can see it.

A federated early-warning grid for India's primary health network. Google's **TimesFM** forecasts demand for every facility–drug pair, the detector spots an outbreak surge days before a shelf empties, and the planner moves medicine across district lines before it does — without creating a stock-out anywhere else.

Build with AI: [Code for Communities](#)

Problem statement 03 – Smart Health & Supply Chain Resilience

Live: aarogya-grid-215071922486.asia-south1.run.app

2,824

FACILITIES TRACKED

81,104

STOCK POSITIONS

5,578

DISPATCH ORDERS PLANNED

16

FEDERATED STATE NODES

THE PROBLEM

A sub-centre runs out of anti-snake venom in monsoon. The vials exist.

They are ninety minutes away, in a facility that will write them off unused in six weeks. Nobody knows either fact.

Both facilities report into a paper register that reaches the district office weeks later, if at all. India's public health supply chain does not primarily fail on procurement volume — it fails on **visibility**, and on **lateral movement**: the ability to see a shortage forming and move stock sideways before it becomes a stock-out.

5,030

POSITIONS AT ZERO STOCK

4,683

CRITICAL POSITIONS

37.21 Cr

PEOPLE IN CATCHMENT

₹14.52 L

STOCK HEADING TO EXPIRY

Four users, one dataset — and a loop that comes back.

ANM SUB-CENTRE	Speaks one sentence into a phone instead of filling a register. Told at once if what she reported looks wrong.
Storekeeper PHC / CHC	Gets an order naming a batch, a quantity, an expiry and a distance. Nothing to interpret.
District Health Officer DISTRICT	Asks in their own language, gets a grounded answer with its working shown, and signs what can move.
State & ministry NATIONAL	Sees which districts fail structurally rather than unluckily — and gets a national demand model without any state handing over its facility rows.

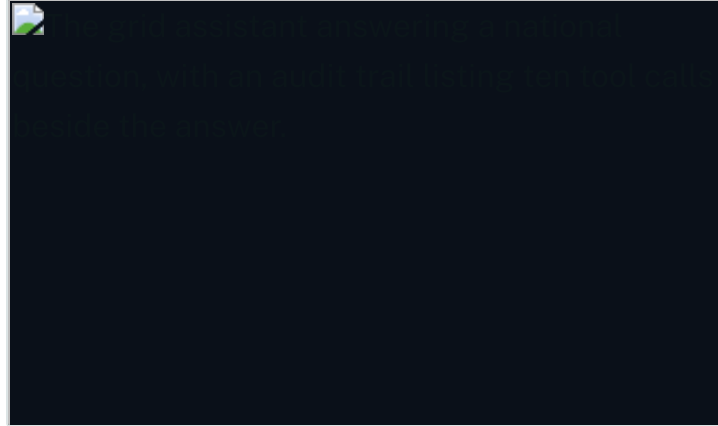
- OBSERVE** Gemini transcribes spoken Hindi or reads a photographed register. **A human confirms** before anything is committed.
- PREDICT** TimesFM forecasts 21 days ahead; a Monte Carlo over the lead time turns that into a stock-out probability and an expected shortfall.
- RECOMMEND** A dispatch order: named batch, expiry date, distance, price — bounded by what the batch physically holds.
- EXECUTE** approve → dispatch → receive, on an append-only audit trail. A short receipt is a **variance**, not a rounding.
- MONITOR** Re-scored server-side in **11 ms** and pushed over SSE. Survives a reload and a killed container.

Most systems of this shape stop at **Recommend**. Nothing afterwards comes back, so nobody learns that 14 vials were sent and **11 arrived**. Here the receiver's risk recovers by what turned up.

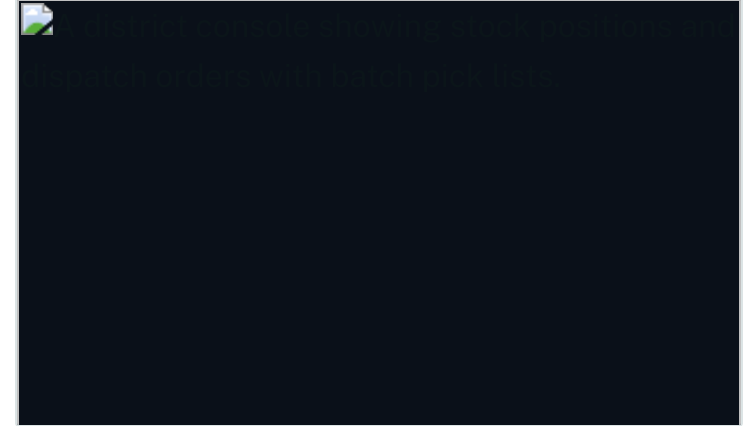
The national view, the assistant's working, and one district's orders.



1 — **/console**. Every bubble is a district; every arc is medicine actually moving.



2 — **Ask the grid**. The audit trail is the product: ten tool calls, every row they read.



3 — **/district**. Batch, expiry, price — and who has to countersign.

Not a dashboard alert. A dispatch order a storekeeper can act on.

GENERATED DISPATCH ORDER · BHAGALPUR → PURNIA, BIHAR

DH Purnia-01 holds **43 bottles** of Ringer Lactate — a **Vital** drug — against a **97%** chance of running short inside its 10-day resupply window. **DH Bhagalpur-01**, in the next district, can spare **110**. Moving them **100.6 km** costs **₹427** and averts an expected **53.3 bottles** of unmet demand, cutting stock-out risk from **97% to 5%**.

pick list: B010-DH001 × 12 (exp 2027-10-28) · B011-DH001 × 98 (exp 2028-04-27) · risk reduction 92 pp · donor left at 0% stock-out risk

Every recommendation names the **specific batches and their expiry dates**, oldest first, and the quantity is bounded by what those batches physically hold. A recommendation a storekeeper cannot fill is not a recommendation. This one crosses a district boundary — the clause the brief adds this edition — so it also carries who is allowed to issue it: **Purnia cannot approve it alone**, and the console disables Approve until Bhagalpur countersigns. **₹427** rather than **₹2,261** because four other orders ride the same vehicle.

TimesFM forecasts. Gemini reads and retrieves. Neither invents a number.

TIMESFM 2.0 All **6,016** district × drug series, **21 days** ahead from a **90-day** context, through BigQuery **AI.FORECAST** — three statements, **0 bytes billed**, none declined.

GEMINI Spoken Hindi and photographed registers into structured entries; and a **function-calling** surface of **12 tools** over the computed state, with the **tool trace on screen**.

DETERMINISTIC Censored Croston-SBA, Monte Carlo risk over each facility's own **7–18 day resupply window**, and the redistribution optimiser. Reproducible from a seed.

NEVER THE MODEL No arithmetic, no drug code, no facility id, no status. The drug is resolved **server-side from its name**, never from a client-supplied id.

CENSORING A ledger records what was **dispensed**, not what was **needed** — a stock-out looks exactly like a quiet week. Fitting naively under-orders for the facilities already failing, so the fit excludes stocked-out periods.

DEMAND CLASS	POSITIONS	TIMESFM	CROSTON	SERVES
Intermittent	24,187	0.896	0.958	TimesFM
Smooth	26,789	0.761	0.771	Croston
Erratic	15,964	0.706	0.696	Croston
Lumpy	7,628	0.804	0.840	Croston

MASE on a **28-day held-out** backtest neither model saw. TimesFM takes a class only where it beats Croston by more than **5%**. It wins intermittent demand by 6.5% and holds **30,535** of **81,104** shipped positions. **We publish where it loses.**

AND THE FIT IS ON CORRECTED HISTORY	BIAS	ERROR
The raw ledger	-3.8%	18.4%
Stock-outs excluded	-1.0%	15.5%

Never MAPE: it divides by the actual, and intermittent demand is full of zeros · full table in docs/forecast-backtest.md

A stock-out is a lagging indicator. We publish how often the warning is wrong.

By the time a block's anti-malarials run short, the malaria has been there a fortnight. **AI.DETECT_ANOMALIES** runs over district consumption and OPD attendance — 6,144 series, 7 statements, **0 bytes billed** — and a warning is a **rule over points**, tuned against **126 injected surges** and scored on 60 candidate rules.

Detection of a 2× 14-day surge	100%
Median lead before the first shelf empties	4.04 days
False alarms per district-week	0.395
Precision	23%

23% precision is not a good number and it is on this slide anyway, next to the 59 rules that failed. Two tighter rules reach 48% and 57% and miss only the four-day lead. Moving the gate after seeing the table would turn every number here into an argument.

The measurement inverted the obvious expectation. Footfall is upstream and nearly blind at district scale: an outbreak doubles one disease, and vector-borne illness is ~9% of a district's OPD, so a 2× surge arrives in total attendance as **×1.09** while the drugs that treat it double. The tuned rule runs on **consumption**.

SIMULATE_OUTBREAK · PURNIA, VECTOR-BORNE ×2

14 of 76 surge needs servable at routine valuation, **35 at emergency**, for **₹31,906** more transport. An outbreak has to raise the **price of a stock-out**, not only demand — a ministry dial, reported next to its effect rather than baked in.

States keep their data and share only models. Here is exactly what crossed.

25,184

NUMBERS SHARED, 16 STATES

10,82,880

ROWS THAT STAYED PUT

- facility rows
- stock quantities
- patient records
- district identifiers

Each state fits its own model and publishes a monthly demand multiplier per catalogue item, its standard error, and one vacancy rate per cadre. Pooled by **random effects with the between-state variance estimated from the nodes**, so a state keeps its own estimate exactly to the extent its data earns it.

Checkable, not assertable. GET /api/federated/10 returns Bihar's node byte for byte; the index publishes the SHA-256 of all sixteen. **npm test** sweeps every file for facility ids, district codes and batch numbers — after first proving on **five poisoned copies** that the sweep can still see one.

A STATE JOINS WITH 30 DAYS	SCALED MAE
No seasonality at all	0.2589
Its own fit, alone	0.2555
Shrunk toward the national prior	0.1574
Its own fit on all 180 days (ceiling)	0.1513

38.4% closer to observed demand than forecasting alone, measured **leave-one-state-out** — the prior it is offered is pooled from the other fifteen states only.

Limitation, stated before anyone finds it: one seeded simulator generates all sixteen states, so the pooling weights are a demonstration of a mechanism rather than a finding about Indian states. It is on the console panel too.

It does not pay for itself in cash. It pays in shortfall, and we lead with the sign.

NATIONAL PLAN	VALUE
Waste averted	+ ₹3.8 L
One dedicated vehicle per order	— ₹70.9 L
Transport cost, consolidated	— ₹29.8 L
Net cash position	— ₹26.0 L
Shortfall averted	5,42,644 units

It breaks even at ₹4.79 per averted unit of unmet demand. Whether a dose of a Vital medicine reaching a patient is worth that is a policy judgement, not an engineering one — so the shortage penalty is an explicit parameter, not something folded into a headline.

Visibility was not the binding constraint. A truck per tablet was. Of 10,353 needs the planner declined, 95.0% failed the benefit/cost gate — the medicine existed, a median 26.8 km away, and the trip cost more than the medicine. Only 9 failed for want of stock anywhere.

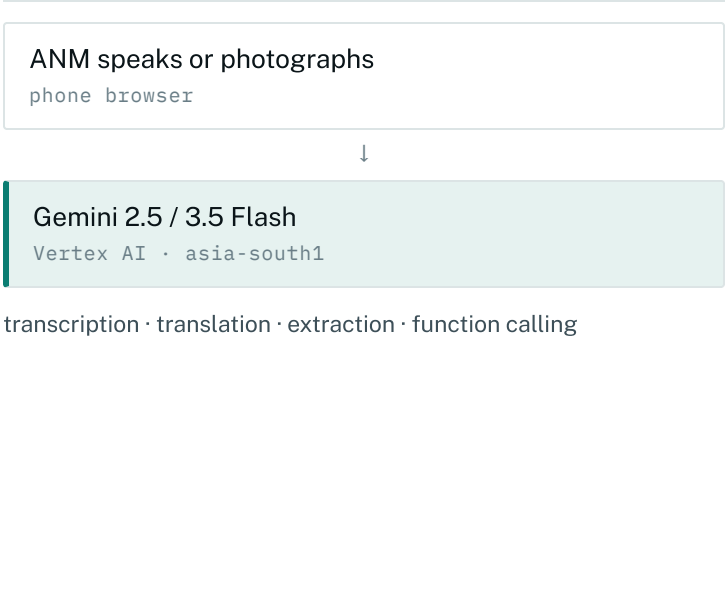
NATIONAL PLAN	BEFORE	AFTER
Dispatch orders	2,798	5,578
Vehicle trips	2,798	2,078
Crossing a district	0	615 trips

3,008 of those orders could not justify a vehicle alone. They ride a trip already going, for the price of handling — stock-outs that were being refused for a truck nobody needed to hire.

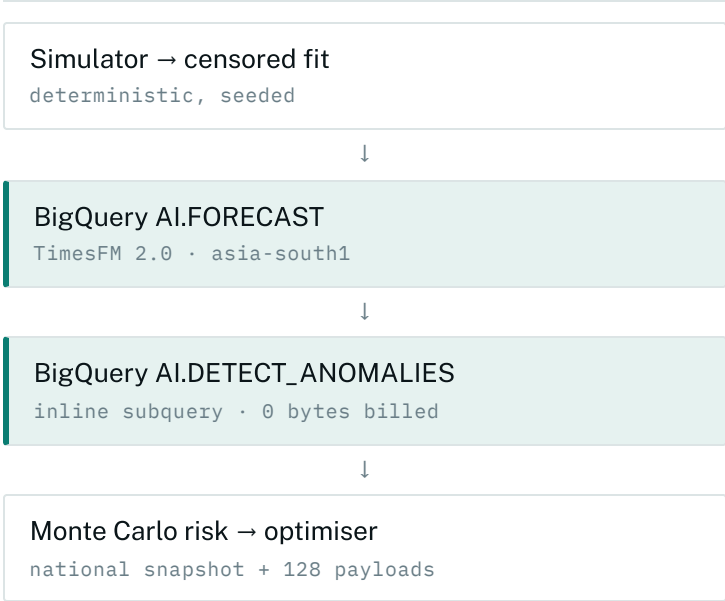
Summed from the 128 shipped district payloads · and no donor is pushed past 10% stock-out risk to get there: 196 donor positions audited, worst 3.0%

One region, one container, four Google services, nothing else to operate.

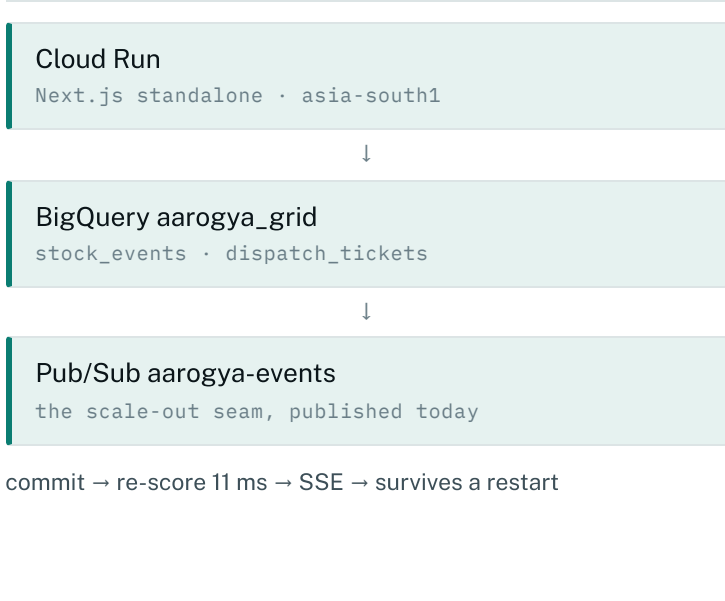
FIELD



BATCH, NIGHTLY



SERVE & REMEMBER



Every inference call stays in asia-south1 · the only recurring cost is the Cloud Run min-instance floor (~₹1,200-1,800/month); the forecast queries process 0 bytes and are free at this volume

Stated plainly, because the honest answer is the stronger one.

REAL 128 districts with their LGD/Census state codes · **Census 2011 district populations** · the NLEM drug catalogue, forms, strengths and VED classes · IPHS tier, bed and staffing norms · an NFHS-5 state indicator · the national boundary from DataMeet's composite file.

SIMULATED Facility-level stock, consumption, batches and expiries · bed occupancy · staff attendance · OPD footfall. Labelled **SIMULATED FACILITY DATA** in the console header and on the landing page, and itemised in NOTICE.

Why simulated? Because the absence of this data *is* the problem. If per-facility stock across India's PHC network were available as an extract, the shortage would already be visible. **Three adapter functions** stand between this and a DVDMS or e-Aushadhi extract; nothing downstream changes.

OWN IT No authentication. Writes are server-side only; the actor is recorded as **actor_claimed**, which is what it is.

OWN IT **--max-instances=1** during judging: the live overlay is in-process. The scale-out step is a subscriber on a topic we already publish to, deliberately not built.

OWN IT The federated τ^2 is **synthetic** — one simulator behind sixteen states.

OWN IT The batch is a script, not a scheduled job. One machine, run by hand.

Nothing on any surface is typed by a human. A drift guard re-derives every figure in this deck, the README and the defence pack from the shipped payloads on every **npm test**. It has caught a slide quoting a dispatch order that no longer existed.

One state, eight weeks, two metrics — then the same code, more rows.

- WEEK 1–2** Point the three adapters at one state's DVDMS/e-Aushadhi extract. Positioned as a **decision layer over** DVDMS, e-Aushadhi and IHIP — not a replacement.
- WEEK 3–6** Two districts live: the ANM capture path, the DHO console, dispatch tickets in the district office.
- WEEK 7–8** Measure **two** things and nothing else: **stock-out days prevented**, and **time from alert to action**.
- REACH** **174** district-to-district corridors carry medicine in the shipped plan, **29 cross-state corridors** among them — each one classified, and each one refusing to be approved by a district officer alone.
- SCALE** 128 districts in **190–240 s** on one laptop. District-parallel with no shared state; clustering colours into **9 concurrent rounds**, not 128 tasks. All-India extrapolates to **~25 min** at the slow end of the measured rate.

BRICS portability. The core is country-agnostic: forecasting, censored fitting, risk, and a distance-and-price optimiser. What is country-specific is a **medicines list**, **facility norms**, a **currency** and a **language pack** — four adapters, not a rewrite.

The early-warning feed already leaves in a schema nobody has to be Indian to read: an area has a code, a named code system, a name and a population; a signal has a hazard class from a fixed list, an observed value, an expected range and a confidence. Every Indian identifier travels in an optional block a consumer can drop, and **npm test** strips that block and re-validates.

WHAT WE WOULD BUILD NEXT, IN ORDER

A Pub/Sub subscriber so the live overlay survives more than one instance · a real IDSP/IHIP bulletin ingest · per-state backtests once two states have real history · and the thing a ministry will ask for first: **procurement forecasting**, which is the same demand model pointed a quarter further out.